

MILESTONE 5

# AI-Based Forest Cover Change Detection & Monitoring System

*Model Evaluation & Results Analysis*

# Model & Pipeline Recap

## Architecture Flow

**T1 (2021) + T2 (2023)** *6-ch input patches, 256×256*

**Shared Siamese Encoder** *4× DoubleConv blocks → 512ch*

**ASPP Bottleneck** *Dilation rates: 1, 6, 12, 18*

**U-Net Decoder** *Skip connections + UpBlocks*

**Binary Change Mask** *Forest loss detection output*

## Training Configuration

**Best Model** Enhanced SiameseUNet-ASPP

**Loss Function**  $0.7 \times \text{Dice} + 0.3 \times \text{Focal} (\alpha=0.8, \gamma=2)$

**Optimizer** Adam,  $\text{lr} = 1 \times 10^{-4}$

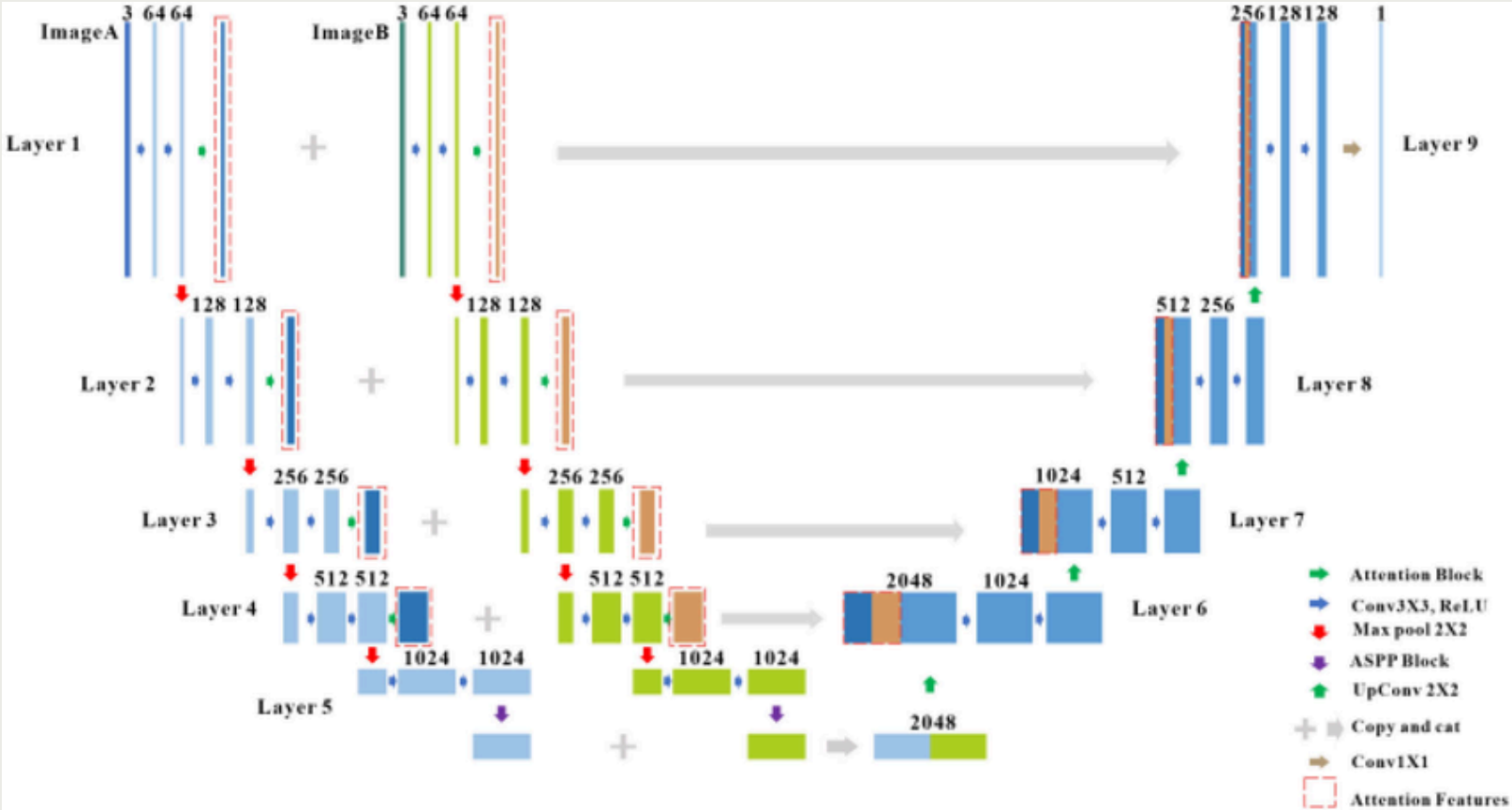
**Training** AMP (float16/float32), T4 GPU

**Scheduler** ReduceLROnPlateau, patience=2

**Batch / Epochs** Batch=16, Max=50, Early Stop=5

**Augmentation** H/V flips + 90° rotations

# Model architecture



# Change ratio distribution

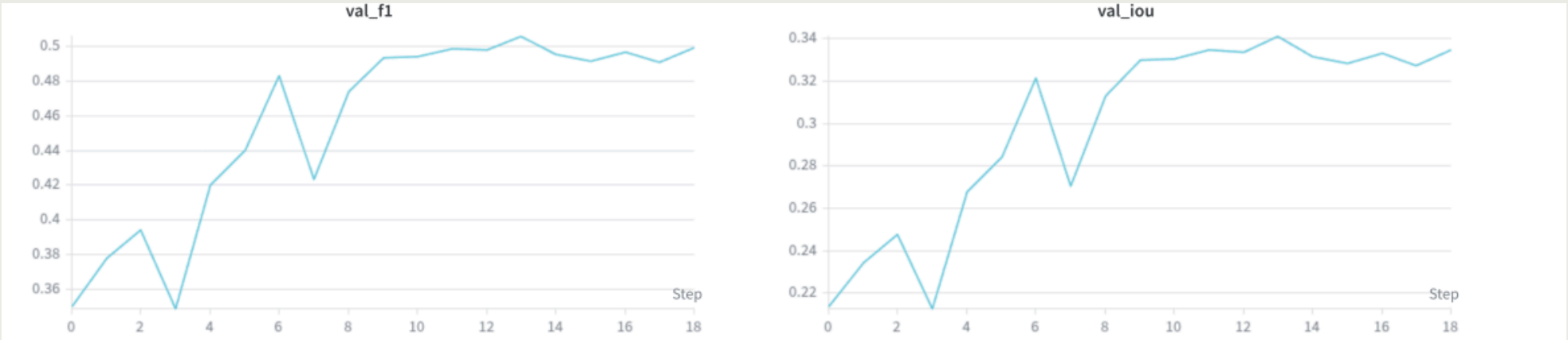
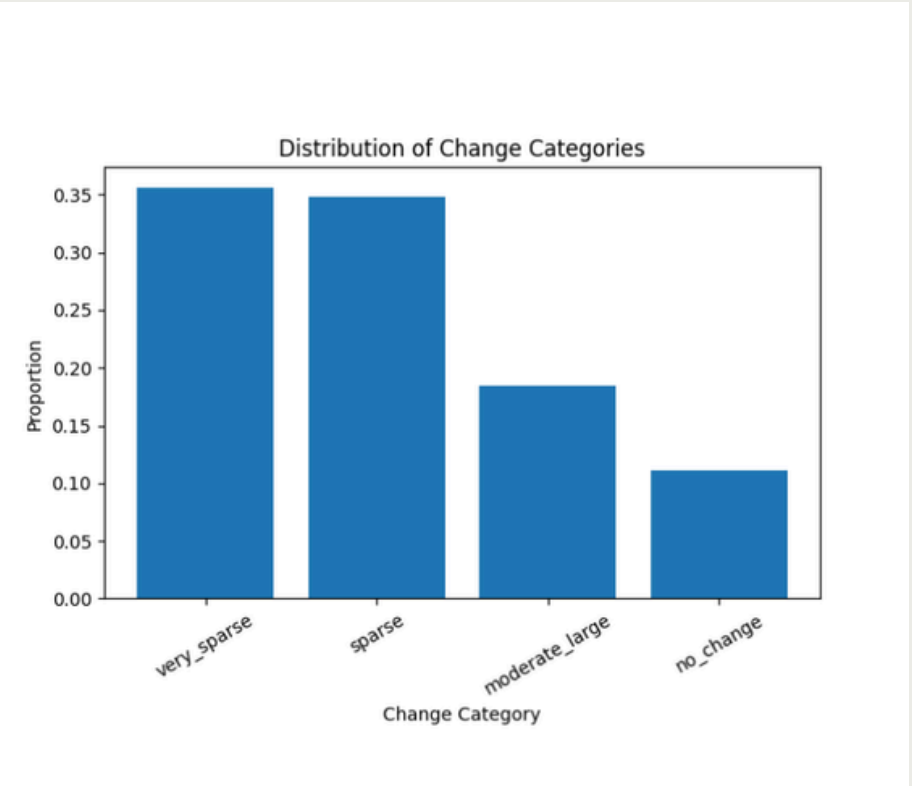


Figure 6: \*

(a) Validation F1 Score

Figure 7: \*

(b) Validation IoU

Figure 8: Validation F1 and IoU curves for the baseline SiameseUNet\_ASPP model across 19 training epochs.

# Evaluation Setup & Metrics



## Dataset & Environment

- 7,500 bi-temporal 256×256 patches — Meghalaya & Nagaland
- Test Set: 867 held-out samples (15% stratified split)
- Regions: Sparse, Very Sparse, Moderate-Large, No Change
- Hardware: NVIDIA T4 GPU (15GB) on Kaggle
- Framework: PyTorch + AMP (float16/float32)



## Performance Metrics (Why?)

- F1-Score → Primary; handles severe class imbalance (1.2% positives)
- IoU (Jaccard) → Stricter overlap, used for model checkpointing
- ROC-AUC → Probability-space discrimination ability
- PR-AUC → Performance under class imbalance
- Precision & Recall → False alarm vs. miss-detection trade-off

**Threshold Strategy:** Optimal threshold searched over [0.05–0.55] → Best = 0.30 (well below 0.5). Class imbalance shifts optimal threshold downward — model assigns low confidence to forest loss predictions.

## Change Category Distribution in Test Set

No Change

**97 samples** 11%

Very Sparse  
(<0.5%)

**308 samples** 36%

Sparse  
(0.5–2%)

**301 samples** 35%

Moderate  
Large (>2%)

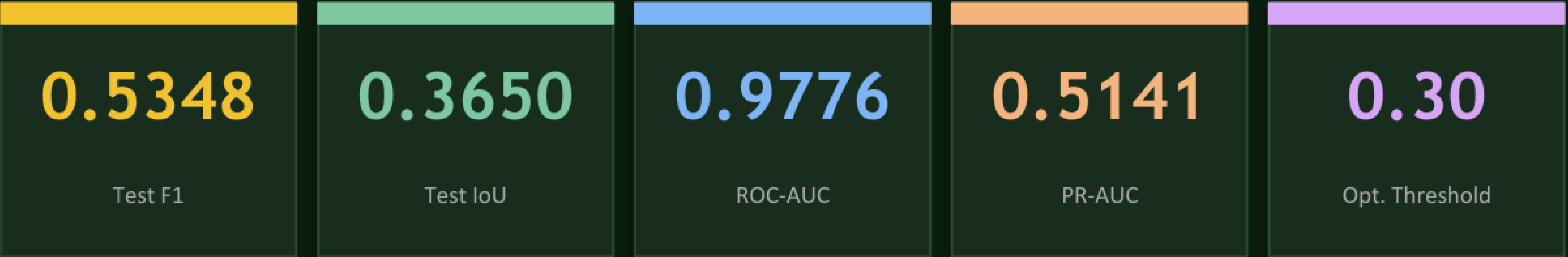
**161 samples** 19%

# Quantitative Results

Model Comparison & Test Set Performance

| Model                          | Best F1 | Best IoU | Best Epoch | Total Epochs |
|--------------------------------|---------|----------|------------|--------------|
| Baseline (BCE+Dice)            | 0.5060  | 0.3413   | 14         | 19           |
| ★ Enhanced (Dice+Focal + Aug.) | 0.5189  | 0.3526   | 23         | 30           |
| Pretrained ResNet-34           | 0.4625  | 0.3026   | 12         | 17           |
| SiameseUNet V2 (Attn+CF+DS)    | 0.4869  | 0.3240   | 12         | 17           |

Best Model — Test Set Performance



# Evaluation Plots & Per-Category Analysis

## Key Evaluation Plots

### PR Curve

AUC = 0.5141 | Operating point @ threshold 0.30

### ROC Curve

AUC = 0.9776 | Strong discriminative ability in prob. space

### Confusion Matrix

TN=55.7M FP=447K | FN=269K TP=412K

### Threshold Plot

F1 & IoU plateau at [0.25–0.50] | Model robust to moderate threshold variation

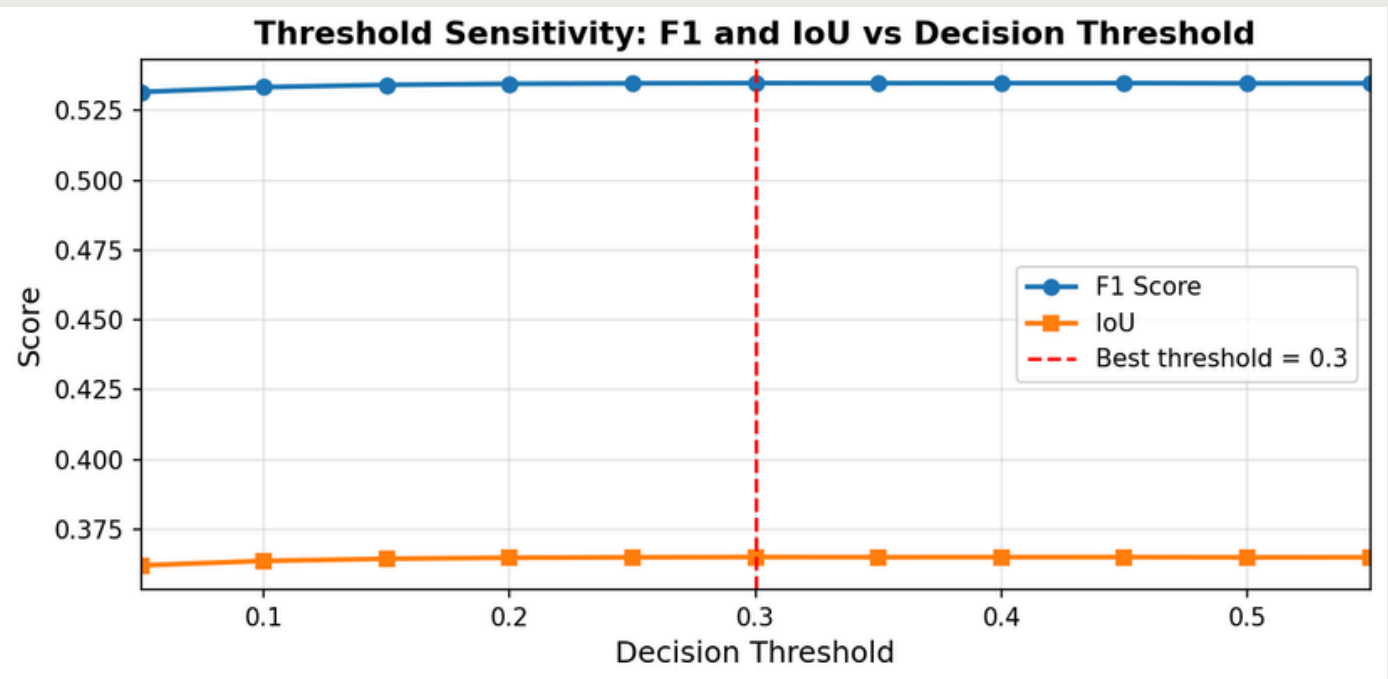
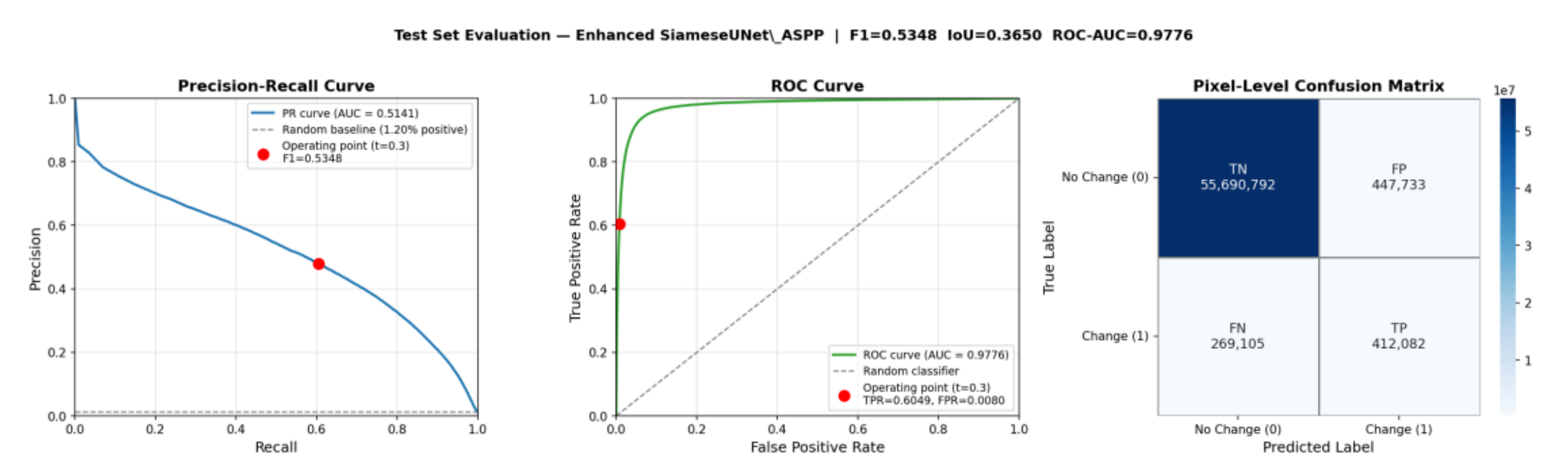
🔍 **Gap Analysis:** ROC-AUC (0.977) vs PR-AUC (0.514) — model ranks positives well in probability space, but cannot achieve high precision and recall simultaneously at any threshold due to 1.20% positive pixel rate.

## Per-Category Performance (threshold = 0.30)

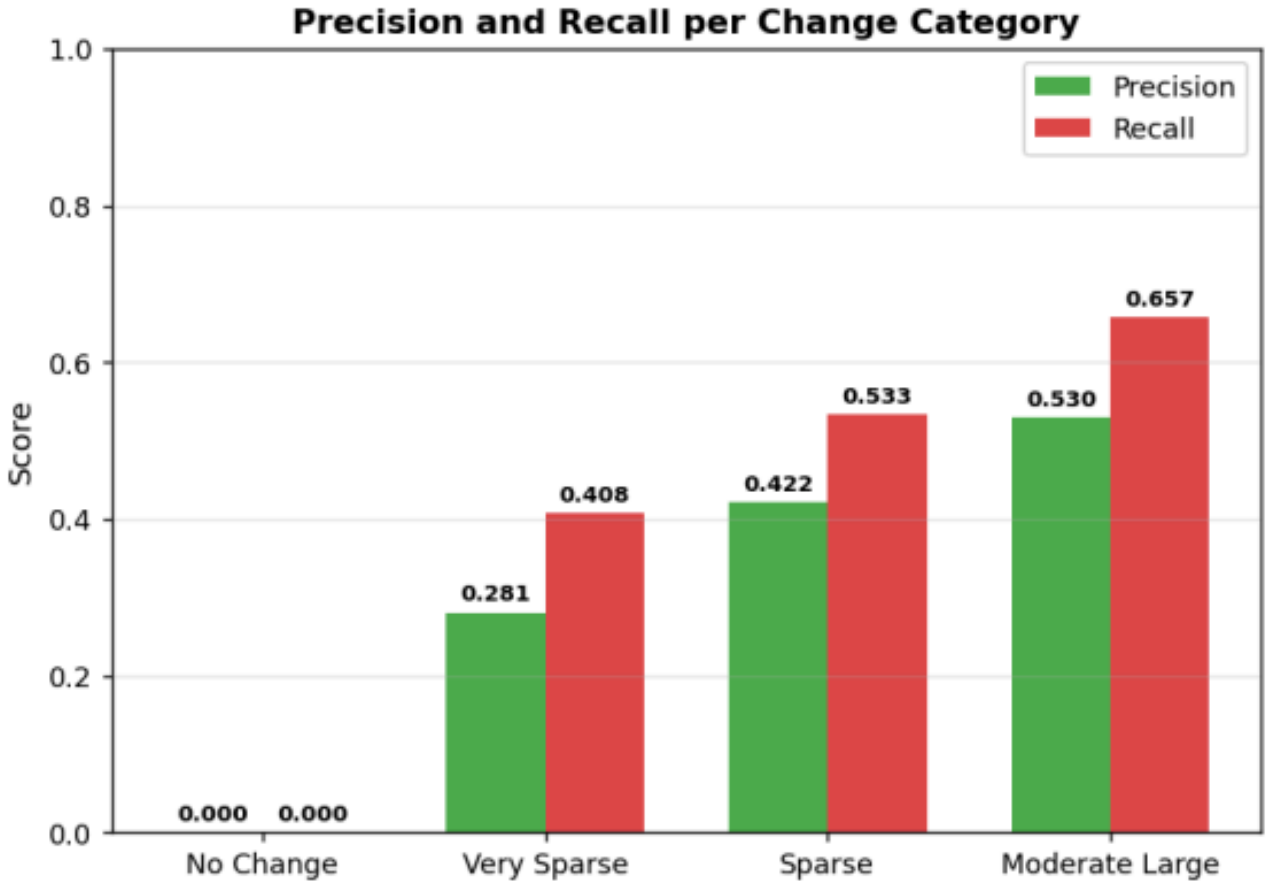
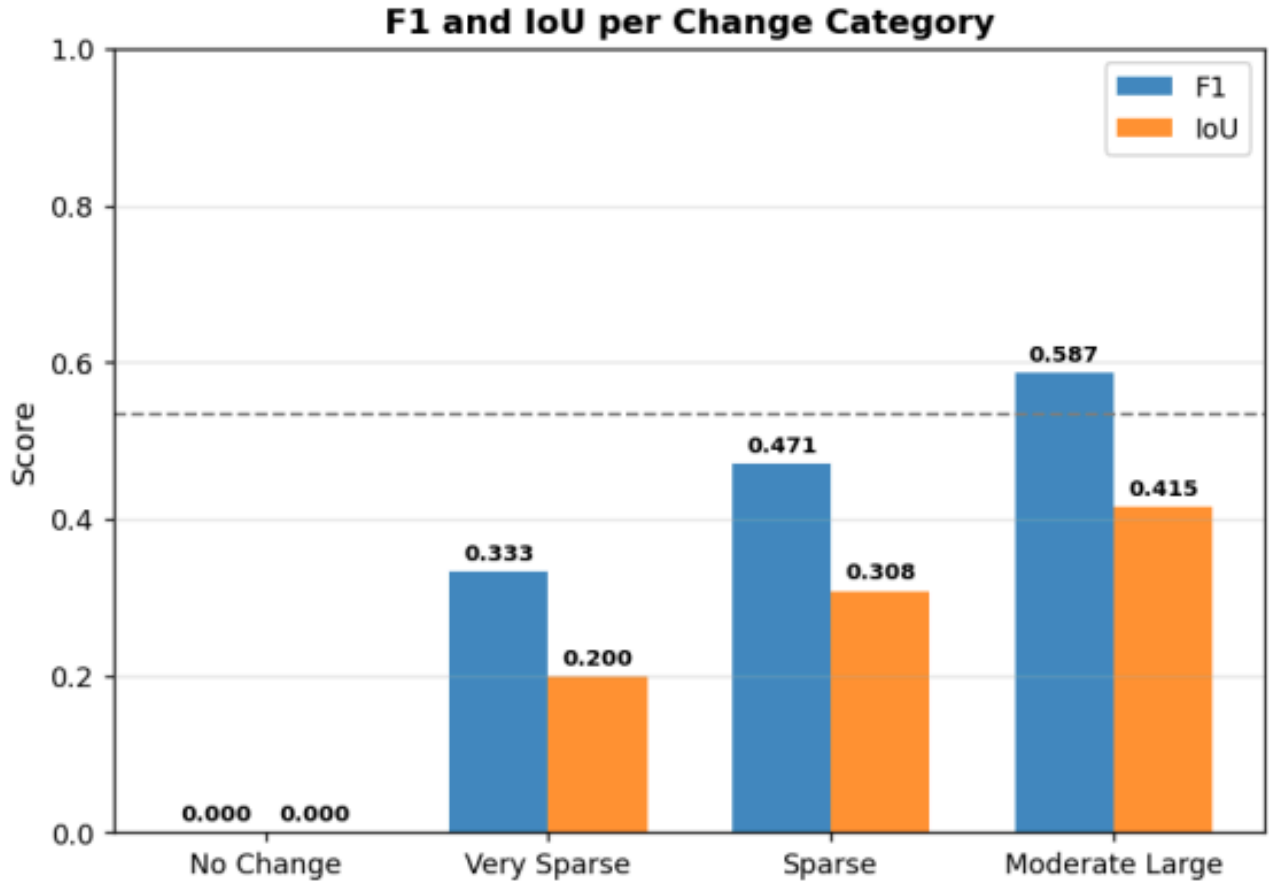
| Category             | F1    | IoU   | Recall | Prec. |
|----------------------|-------|-------|--------|-------|
| No Change (97)       | 0.000 | 0.000 | 0.000  | 0.000 |
| Very Sparse (308)    | 0.333 | 0.200 | 0.408  | 0.281 |
| Sparse (301)         | 0.471 | 0.308 | 0.534  | 0.422 |
| Moderate-Large (161) | 0.587 | 0.415 | 0.657  | 0.530 |
| Overall (867)        | 0.535 | 0.365 | 0.605  | 0.479 |

🔍 Performance scales monotonically with change intensity — moderate-large patches achieve highest scores across all metrics (F1=0.587, IoU=0.415)

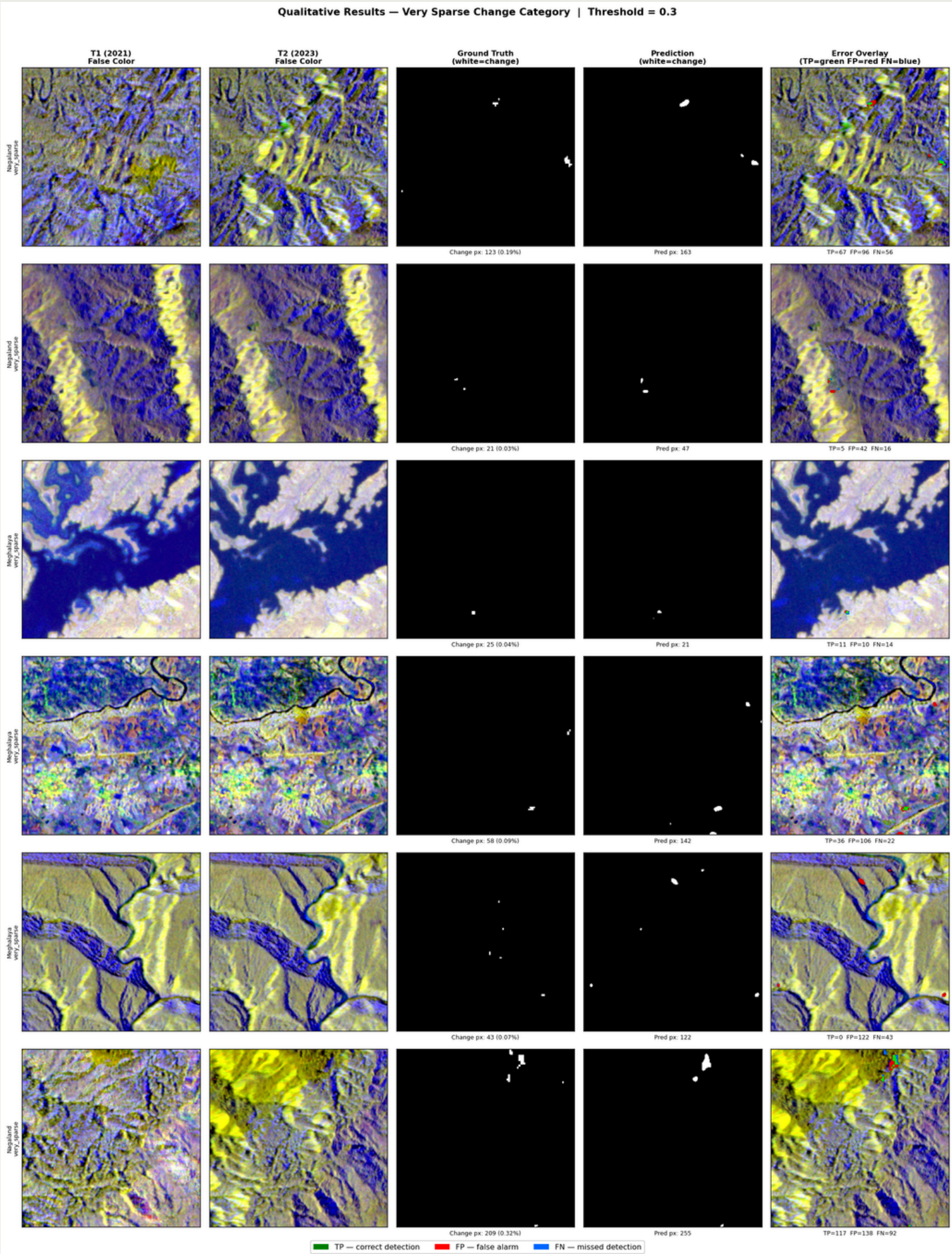
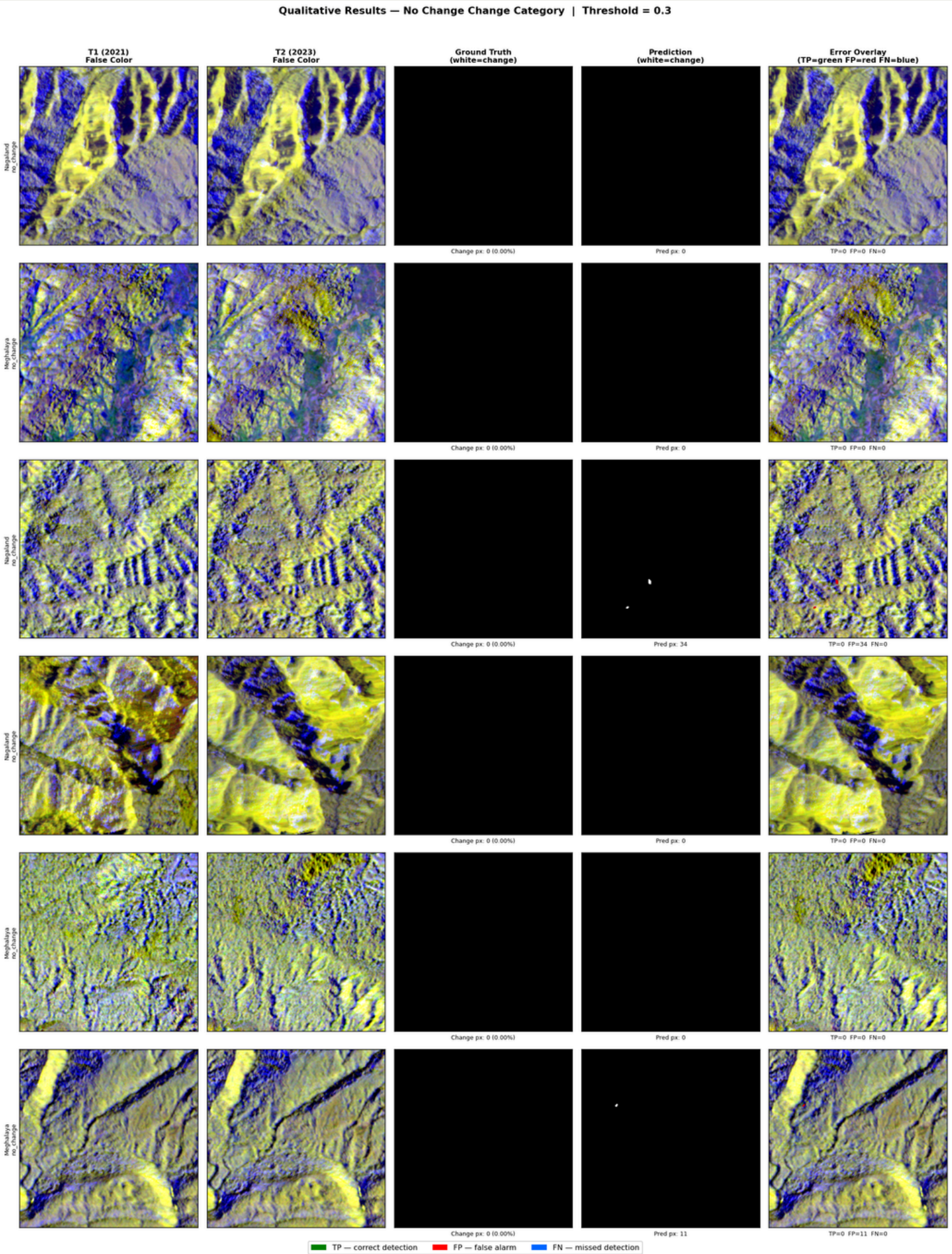
# Graphical representation



**Per-Category Performance — Enhanced SiameseUNet\\_ASPP | Threshold=0.3**

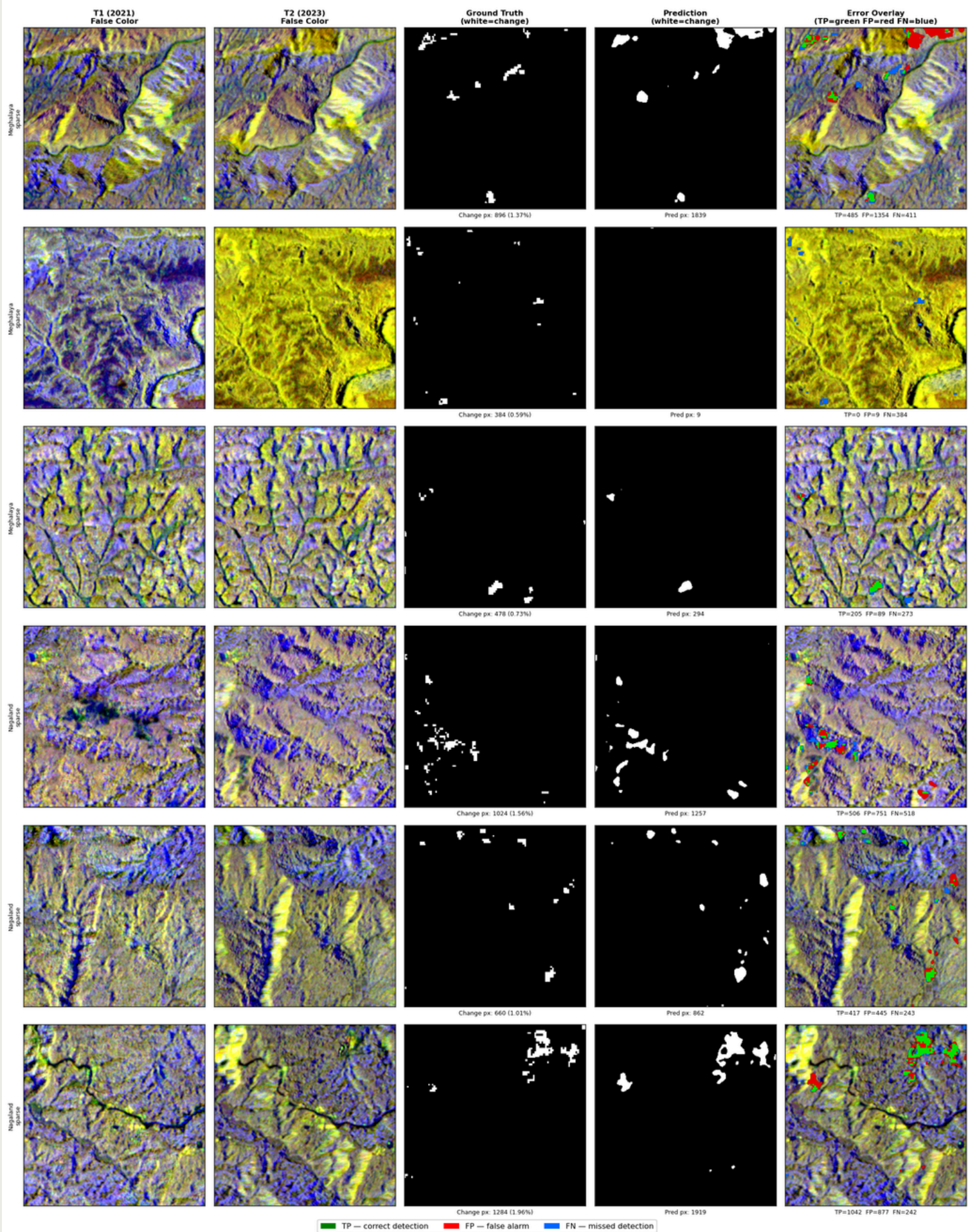


# Output visualization

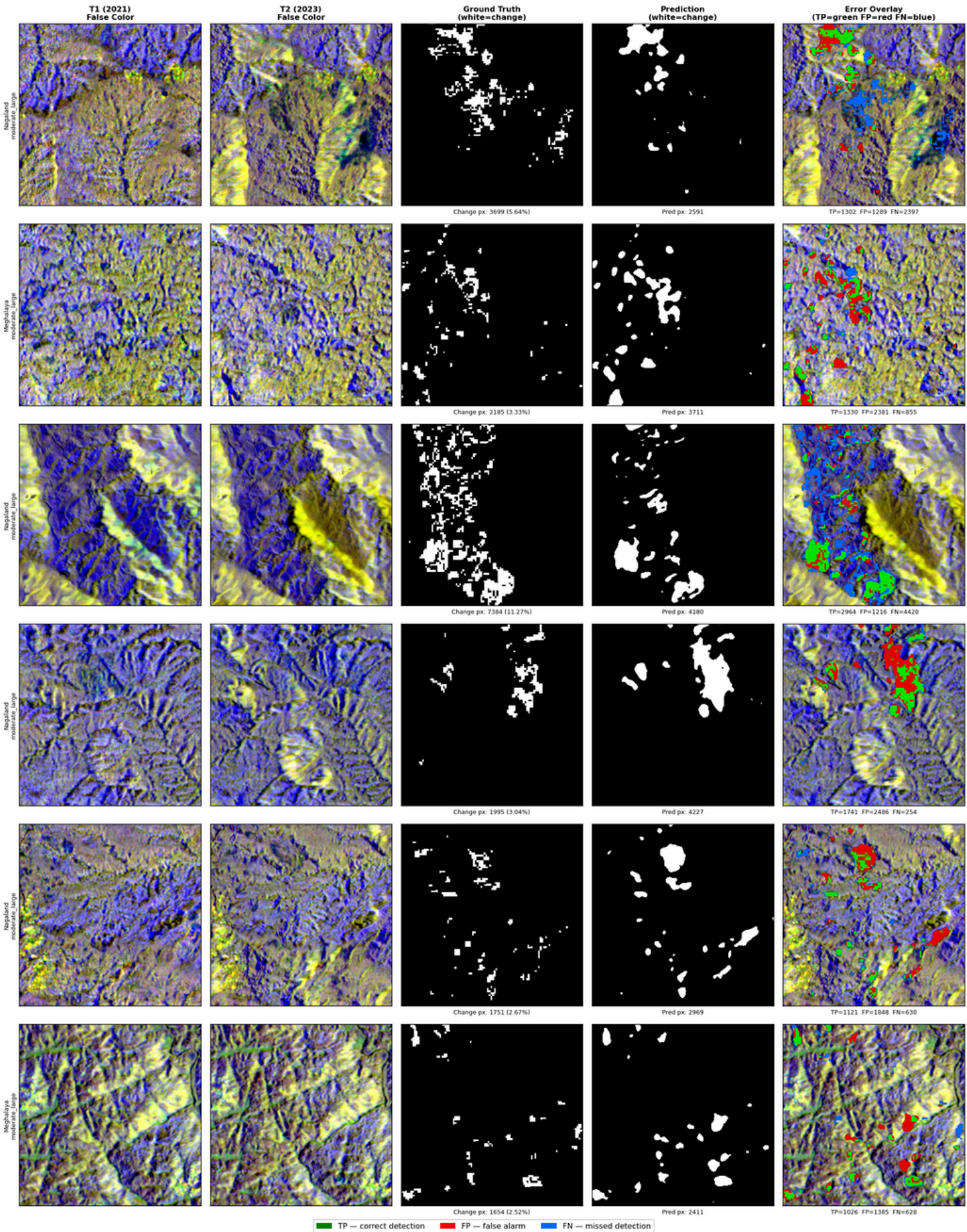


# Output visualization

Qualitative Results — Sparse Change Category | Threshold = 0.3



Qualitative Results — Moderate Large Change Category | Threshold = 0.3



# Qualitative Results & Error Analysis

Prediction Color Code  True Positive (TP)  False Positive (FP)  False Negative (FN)

## ▣ Moderate-Large Change

**Successfully detects large contiguous deforestation clusters.**

Fragmented boundary pixels missed (FN). False alarms along terrain boundaries & spectral transition zones.

## ▣▣ Sparse Change

**Struggles with small isolated Jhum patches.**

Most ground truth pixels are missed (FN). Successful detections only in clusters with contiguous spatial extent.

## ▣ Very Sparse Change

**Very low detection rate.**

Minimal positive predictions made. High FN rate dominates. Small patches lack enough spatial context for confident classification.

## ▣ No Change Patches

**Near-perfect specificity (0.9997).**

Rare false positives in no-change patches. Model correctly suppresses predictions in background-dominant areas.

# Observations, Limitations & Future Work

## 🔍 Key Observations

- Test F1=0.535, IoU=0.365 — consistent with validation, no significant overfitting
- Training-level improvements (loss design + augmentation) outperformed architectural complexity
- Optimal threshold  $0.30 \ll 0.5 \rightarrow$  model under-confident on positive class
- ROC-AUC (0.977) shows good ranking ability in probability space
- Complex models (ResNet-34, V2) underperformed on this 7,500-sample dataset

## 🔍🔍 Limitations

- Small dataset (7,500 patches) — insufficient to leverage complex architectures
- Label noise from 30m Hansen labels applied to 10m Sentinel imagery (boundary uncertainty)
- No cloud masking for Sentinel-2 monsoon imagery  $\rightarrow$  data gaps
- Two-date approach cannot distinguish permanent deforestation from seasonal Jhum regrowth
- Large gap between ROC-AUC (0.977) and PR-AUC (0.514) — fundamental class imbalance (1.20% positives)

## 🔍 Future Work

- Expand dataset across more geographic regions and temporal windows
- Semi-supervised / self-supervised pretraining on unlabelled satellite imagery
- Purpose-built change detection transformers (e.g., BIT — Bitemporal Image Transformer)
- Integrate cloud masking + multi-date temporal sequences
- Combine best strategies from M5 as starting point for final modelling phase

THANK YOU